CPE412 Pattern Recognition

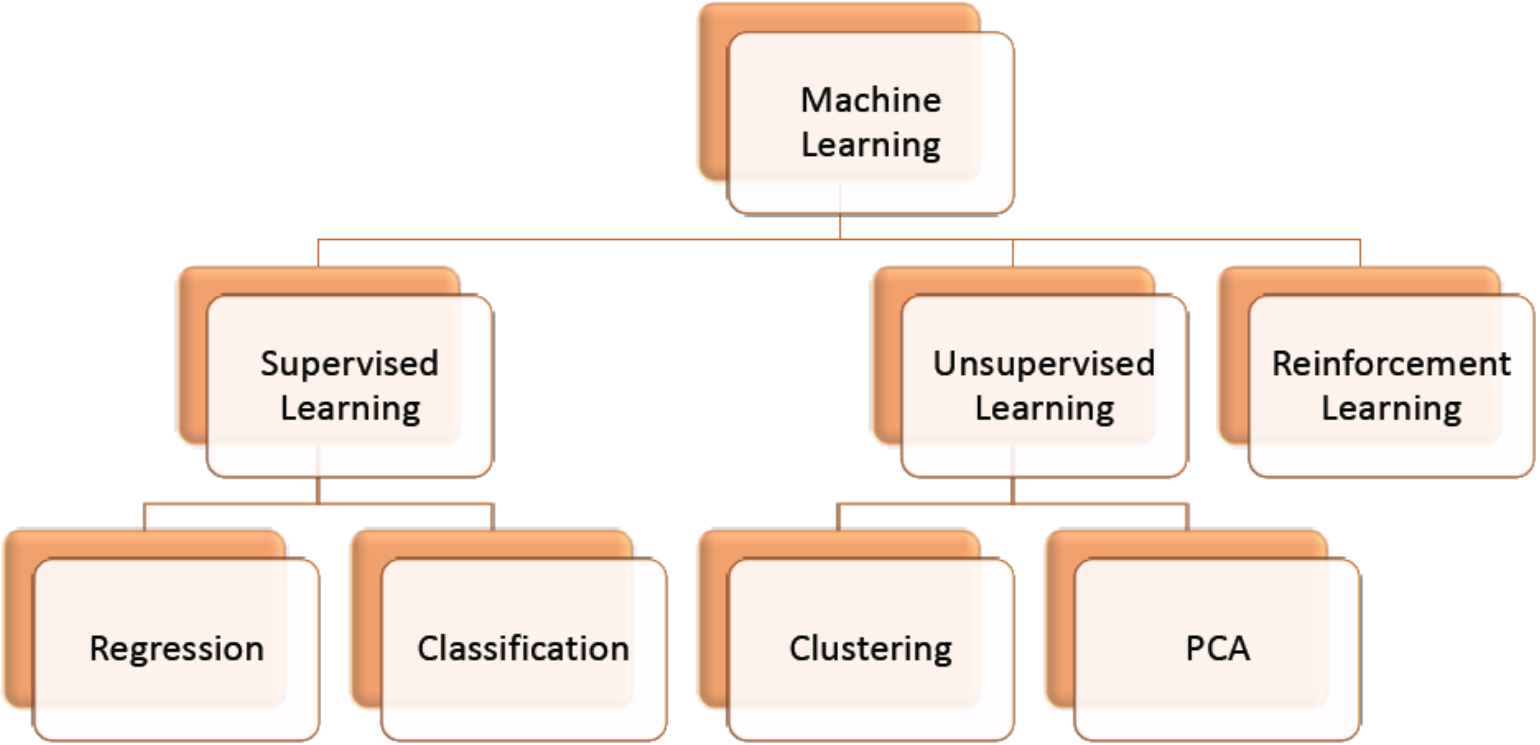
Week 9

Regression and Linear Regression Analysis

Dr. Nehad Ramaha,
Computer Engineering Department
Karabük Universities

The class notes are a compilation and edition from many sources. The instructor does not claim intellectual property or ownership of the lecture notes.

Types of Learning



Techniques

- **Supervised Learning:**
  - kNN (k Nearest Neighbors)
  - Linear Regression
  - Naïve Bayes
  - Logistic Regression
  - Support Vector Machines
  - Random Forests
- **Unsupervised Learning:**
  - Clustering
  - Factor analysis
  - Topic Models

Regression Analysis

- Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables.
- Regression analysis helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed.
- It predicts continuous/real values such as temperature, age, salary, price, etc.

Example

- Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The below list shows the advertisement made by the company in the last 5 years and the corresponding sales:

- Now, the company wants to do the advertisement of $200 in the year 2024 and wants to know the prediction about the sales for this year. So to solve such type of prediction problems in machine learning, we need regression analysis.

Advertisement	Sales
$90	$1000
$120	$1300
$150	$1800
$100	$1200
$130	$1380
$200	??

Regression Analysis

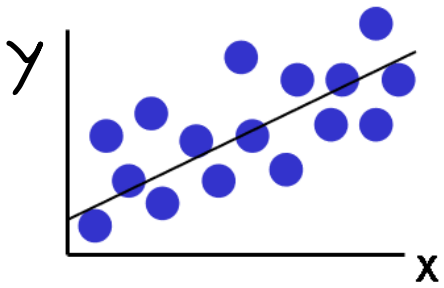
- Regression is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables.
- It is mainly used for prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables.

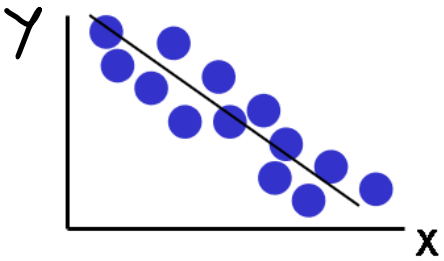
Regression Analysis

- Regression shows a line or curve that passes through all the datapoints on target-predictor graph in such a way that the vertical distance between the datapoints and the regression line is minimum.
- The distance between datapoints and line tells whether a model has captured a strong relationship or not.

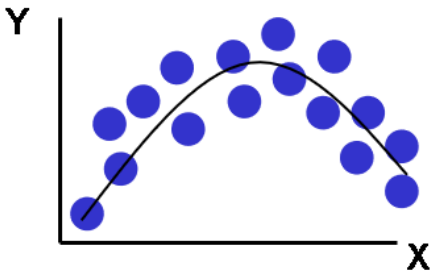
Regression Analysis

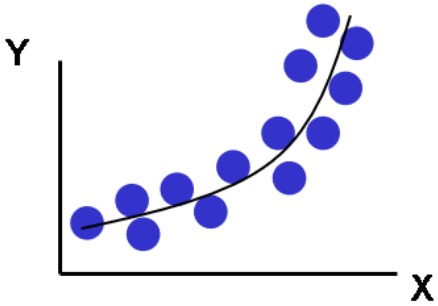
Linear relationships



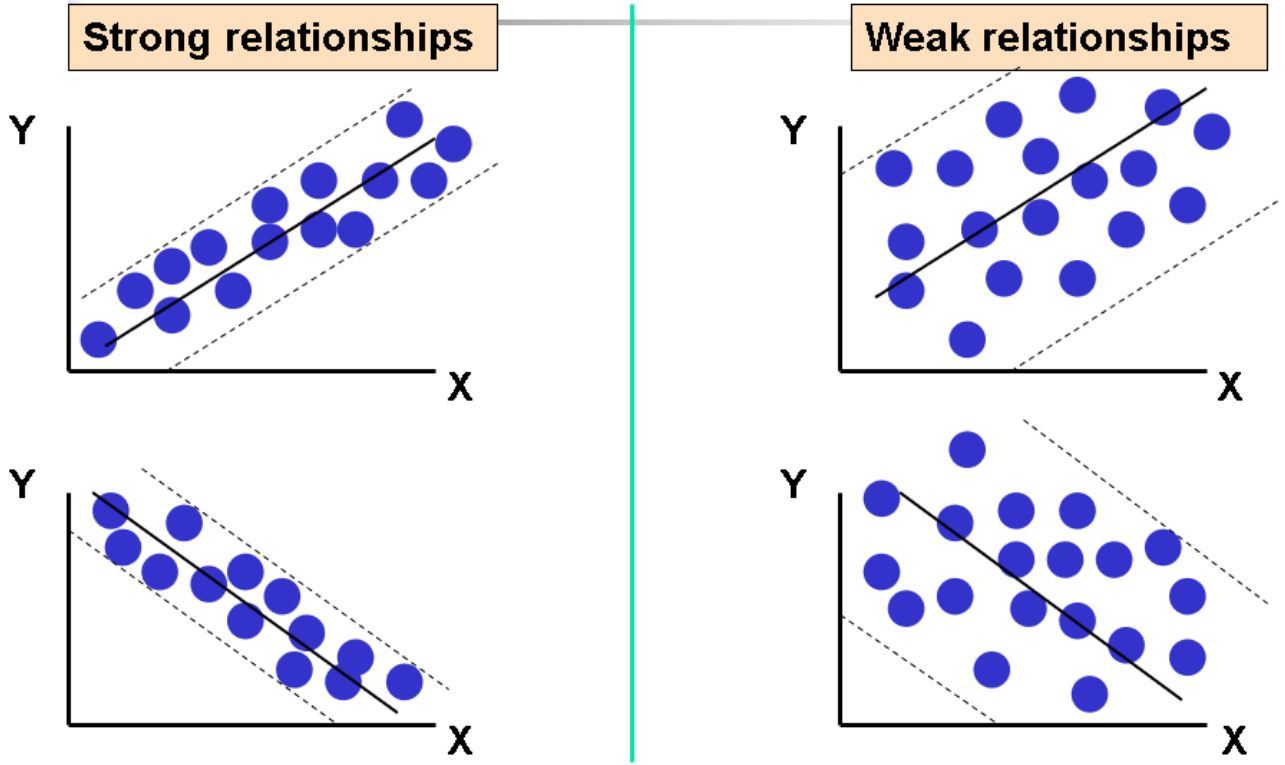


Curvilinear relationships



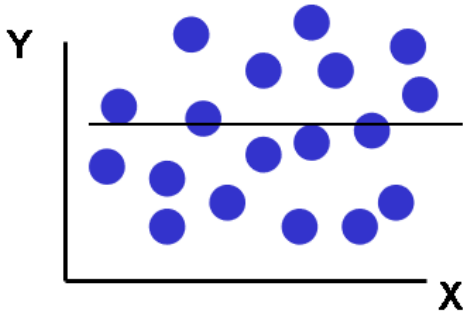


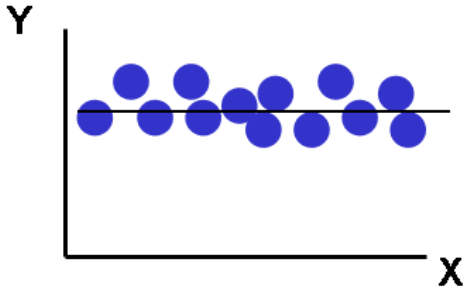
Regression Analysis



Regression Analysis

No relationship





Examples Of Regression

- Some examples of regression can be as:
  - Prediction of rain using temperature and other factors
  - Determining Market trends
  - Prediction of road accidents due to rash driving.

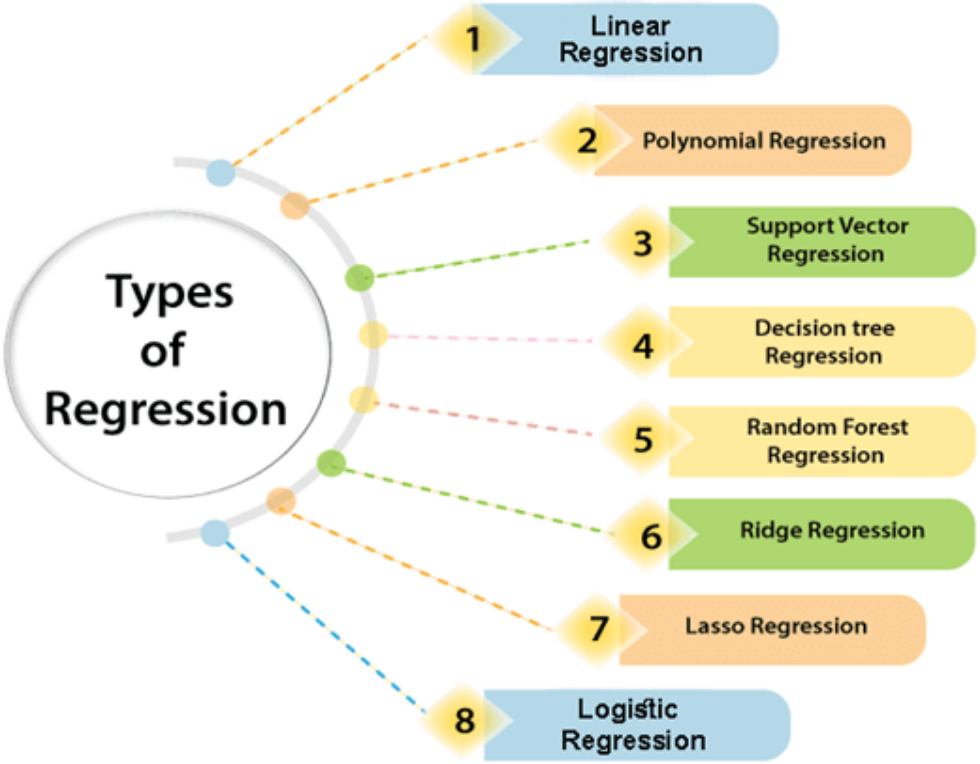
Terminologies Related to the Regression Analysis:

- **Dependent(target) Variable**: The main factor in Regression analysis which we want to predict or understand.
- **Independent(predictor(Variable**: The factors which affect(used to predict) the dependent variables.
- **Outliers**: Outlier is an observation which contains either very low value or very high value in comparison to other observed values. An outlier may hamper the result, so it should be avoided.
- **Multicollinearity**: If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.
- **Underfitting and Overfitting**: If our algorithm works well with the training dataset but not well with test dataset, then such problem is called Overfitting. And if our algorithm does not perform well even with training dataset, then such problem is called underfitting.

Why do we use Regression Analysis?

- Regression analysis helps in the prediction of a continuous variable.
- There are various scenarios in the real world where we need some future predictions such as weather condition, sales prediction, marketing trends, etc., for such case we need some technology which can make predictions more accurately.
- Below are some other reasons for using Regression analysis:
  - Regression estimates the relationship between the target and the independent variable.
  - It is used to find the trends in data.
  - It helps to predict real/continuous values.
  - By performing the regression, we can confidently determine the most important factor, the least important factor, and how each factor is affecting the other factors.

Types of Regression

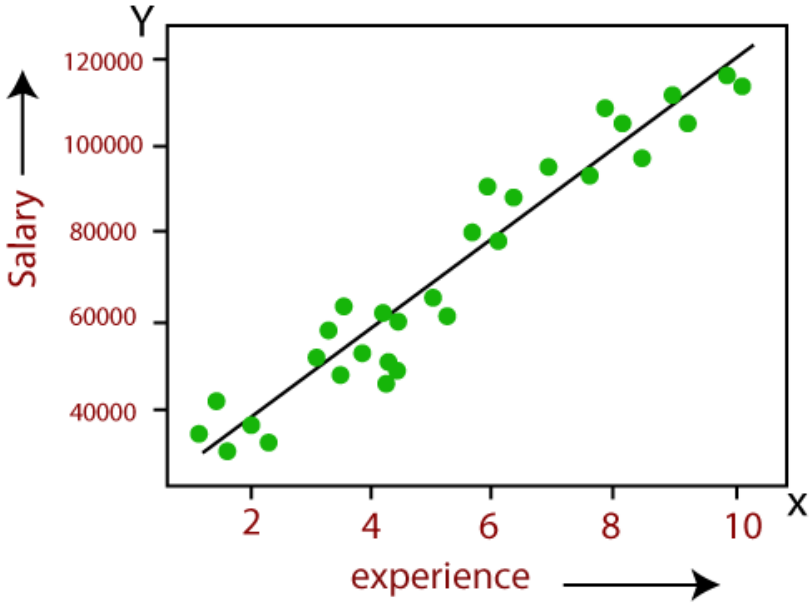


Linear Regression

- Linear regression is a statistical regression method which is used for predictive analysis.
- It is one of the very simple and easy algorithms which works on regression and shows the relationship between the continuous variables.
- Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), hence called linear regression.
- If there is only one input variable (x), then such linear regression is called simple linear regression. And if there is more than one input variable, then such linear regression is called multiple linear regression.

Linear Regression

- The relationship between variables in the linear regression model can be explained using the below image.
- Here we are predicting the salary of an employee on the basis of the year of experience.



Mathematical Equation For Linear Regression

- Linear regression assume the relationship between variables can be modelled through linear equation or an equation of line

Slope

Dependent/Regressed variable ← $y = w_0 + w_1 X$ → Independent/Regressor variable

Intercept

Multiple Regression

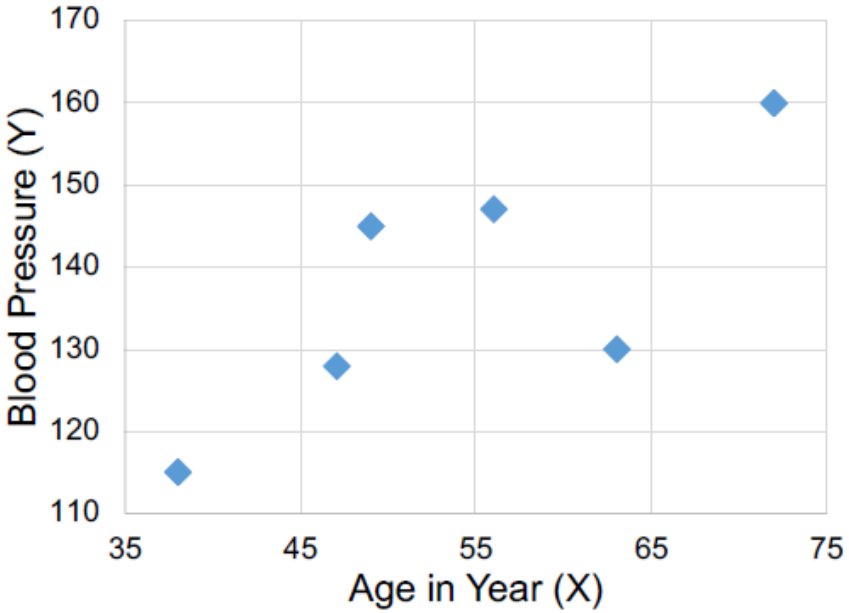
- In Real world a data point has various important attributes and they need to be
- catered to while developing a regression model. (Many independent variables and
- one dependent variable)

$$y = w_0 + w_1x_1 + w_2x_2 + w_3x_3. \dots \dots \dots . \mathrm{w_n x_n}$$

Regression –Problem Formulation

Let you have given with a data:

Age in Years (X)	Blood Pressure (Y)
56	147
49	145
72	160
38	115
63	130
47	128



Regression –Problem Formulation

- For given example the Linear Regression is modeled as:

$$BloodPressure(y) = w_0 + w_1 AgeinYear(X)$$

OR

$$y = w_0 + w_1 X$$ – Equation of line

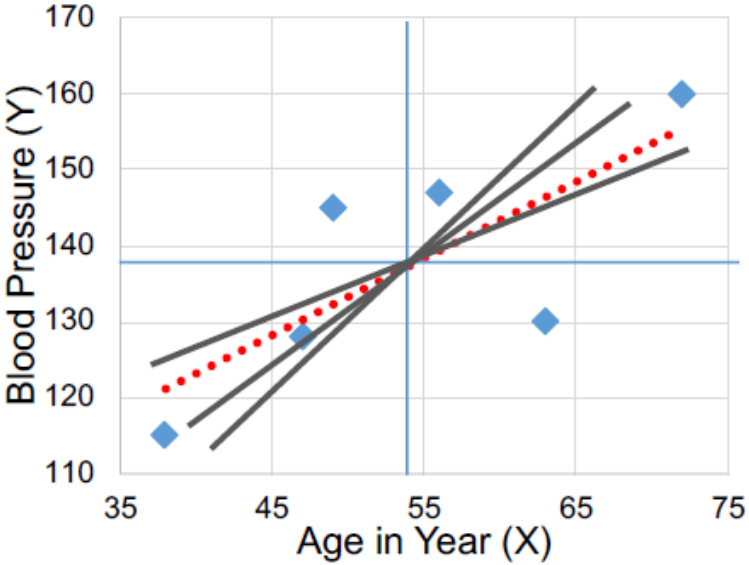
with w_0 *is intercept on Y_axis and* w_1 *is slope of line*

Blood Pressure - **Dependent Variable**

Age in Year - **Independent Variable**

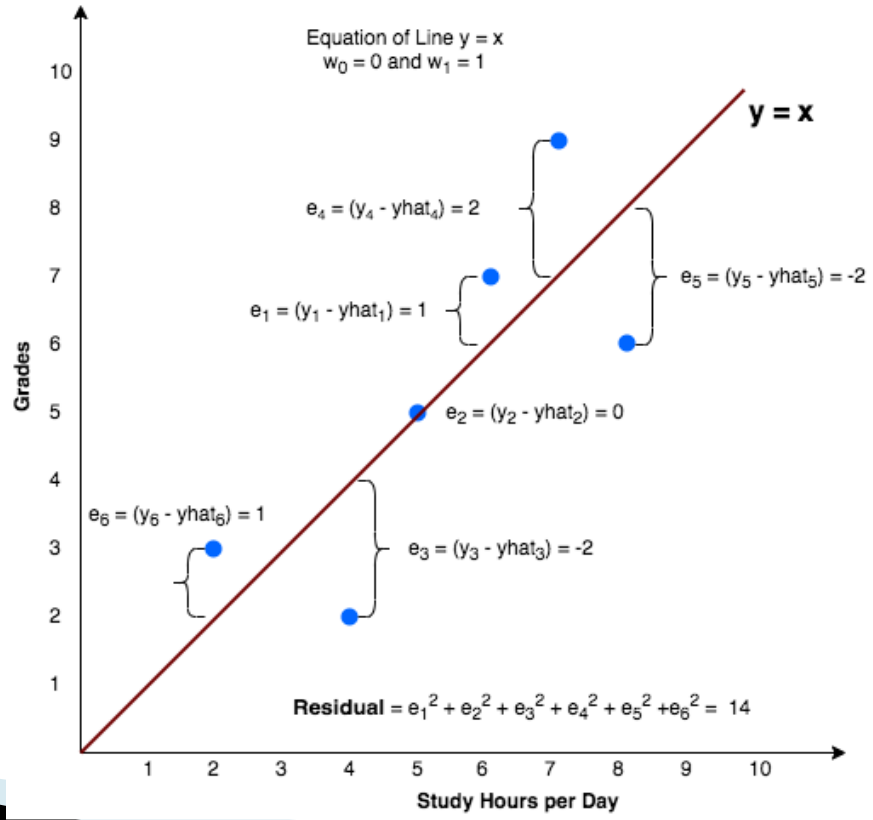
Linear Regression- Best Fit Line

- Regression uses line to show the trend of distribution.
- There can be many lines that try to fit the data points in scatter diagram
- The aim is to find Best fit Line

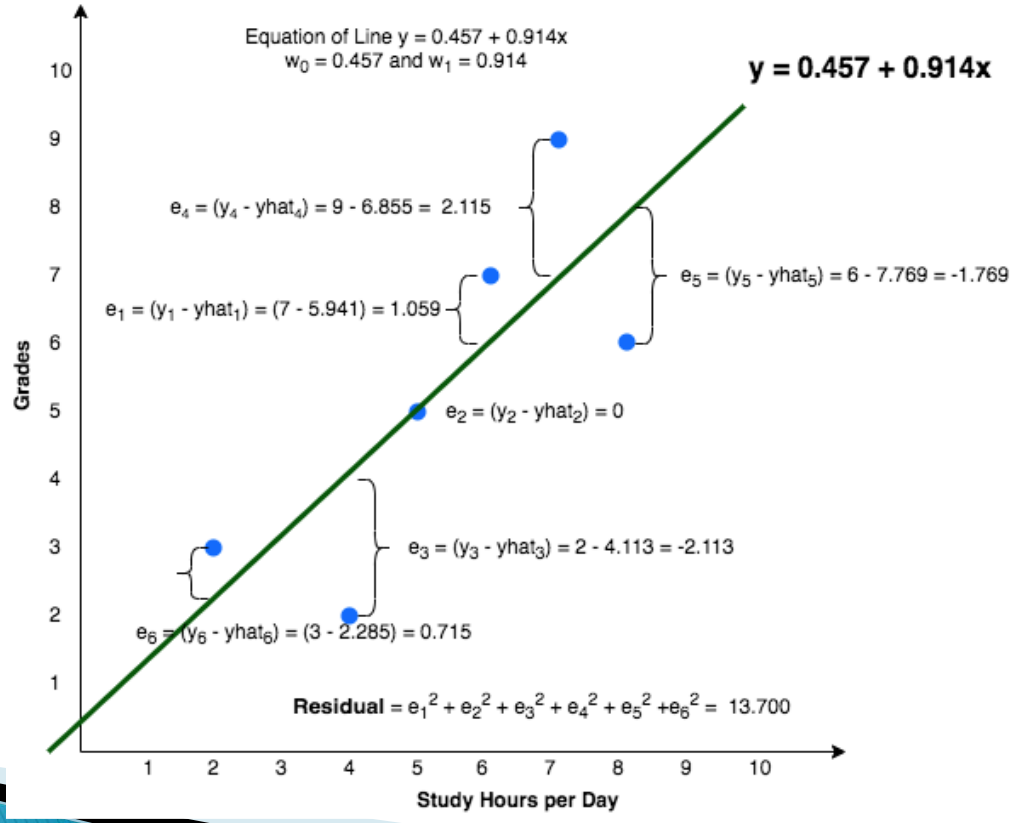


What is Best Fit Line

- Best fit line tries to explain the variance in given data. (minimize the total residual/error)



What is Best Fit Line



Linear Regression- Methods to Get Best

- Least Square
- Gradient Descent

Thanks ☺